

Supported cohomology and counting-preserving inverses for ES divisor boxes

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Abstract

A finite three-term cochain identifies common residues without counting internal cancellation as an arithmetic witness. Its original counting norm has an explicit least-norm quotient. A different, fully marked quotient recovers original divisor states from overlapping split boxes, with inverse-fibre weights and actual relation primitives. These two quotients need not have the same dimension. Iterated inverses agree when the inherited metric is retained, and need not agree when it is reset. The construction is a finite arithmetic instance of the existing SplitZero quotient programme, not a new general cohomology or determinant theorem.

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p/2$, $R = 4a - p$, and $a = \prod q_i^{e_i}$. For a budget $b \geq 0$ define $B_b = \prod [-b_i, b_i]_{\mathbb{Z}}$ and $v(\beta) = \prod q_i^{\beta_i} \pmod{R}$. A channel label c is M or E, with $t_M = -1$, $t_E = -p^{-1}$. The divisor map $\beta \mapsto u = \prod q_i^{e_i + \beta_i}$ is a bijection $B_e \rightarrow \text{Div}(a^2)$. The two gates are $R \mid u + a$ and $R \mid 4u + 1$. These are the inherited ES coordinates [4,5].

Choose $e = f + b$. Retain the two maps, separately at each channel,

$$X = B_f \xrightarrow{\ell} G, \quad Y = B_b \xrightarrow{r_c} G, \quad \ell(\xi) = v(\xi), \quad r_c(\eta) = t_c v(\eta)^{-1}, \quad G = (\mathbb{Z}/R\mathbb{Z})^\times. \quad (1)$$

For any finite maps $X \rightarrow Z \leftarrow Y$, let B_X, B_Y be the free star-difference modules in each nonempty fibre: choose reference x_z , then use $e_x - e_{x_z}$ for every other x . Define

$$B_X \oplus B_Y \hookrightarrow k^X \oplus k^Y \xrightarrow{d} k^Z, \quad d(A, B)_z = \sum_{X_z} A_x - \sum_{Y_z} B_y \quad (2)$$

in degrees $-1, 0, 1$ over any commutative ring k .

Theorem 1 (Cohomology and original counting norm). *Put $I = \ell(X) \cap r(Y)$. The incoming map in (2) is injective, $H^0(2) \simeq k^I$, and $H^1(2) \simeq k^{Z \setminus (\ell(X) \cup r(Y))}$. The degree-zero map sends a cycle to its common fibre sums c_z ; its integral inverse is $c_z(e_{x_z}, e_{y_z})$. For the original counting Gram over $\mathbb{R}$ or $\mathbb{C}$, its canonical representative assigns c_z/m_z to each of the $m_z = |X_z|$ left occurrences and c_z/n_z to each of the $n_z = |Y_z|$ right occurrences. Its quotient squared norm is*

$$\boxed{\sum_{z \in I} (m_z^{-1} + n_z^{-1}) |c_z|^2}. \quad (3)$$

The source, relation and quotient determinants on that fibre are $m_z + n_z$, $m_z n_z$, and $(m_z + n_z)/(m_z n_z)$.

Proof. A vector with zero fibre sum is uniquely $\sum_{x \neq x_z} A_x (e_x - e_{x_z})$. In a cycle, only a common nonempty fibre permits a nonzero sum. Subtracting the displayed reference lift therefore gives exactly the incoming image and its explicit primitive. The image of d is the free coordinate module on the union of the two images, proving H^1 . The stated canonical representative has the same sums and is orthogonal to every star difference; subtracting it proves least norm and the Pythagorean identity. The determinant of a star Gram is its fibre size. Applying the Schur identity to the cycle basis (reference, star differences) gives the displayed determinants. $\square$

The existing SplitZero reconstruction [1,2] retains a support mask and distinguishes external absence τ from $(\lambda, 0)$ in a present fibre. Apply it to (2) at the masks of its two legs. Incoming relations map to their *supported* zero classes, not to absence. The source quotient metric is the same actual-Gram construction as [3,§1]. There is no assertion that the finite counting norm equals the analytic theta source norm.

A residue coarsening $\pi : Z \rightarrow Z'$ induces on H^0 the map $c \mapsto (\sum_{\pi(z)=w} c_z)_w$. Its kernel has star-difference bases inside fibres of $I \rightarrow \pi(I)$; its cokernel is freely indexed by $(\pi\ell(X) \cap \pi r(Y)) \setminus \pi(I)$. Every killed class has the explicit coarser star primitive above. Thus local cohomology classes do not assert one simultaneous original exponent choice.

Theorem 2 (Original arithmetic masses and inverse fibres). *Let $\mathcal{P}_c = \{(\xi, \eta) : \ell(\xi) = r_c(\eta)\}$ and $\mathcal{W}_c = \{\beta \in B_e : v(\beta) = t_c\}$. Addition $q : \mathcal{P}_c \rightarrow \mathcal{W}_c$ is onto, and its complete fibre is*

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi. \quad (4)$$

Let $N(\beta)$ be the product of these interval lengths. For every original rational weight w_β ,

$$\sum_{\beta \in \mathcal{W}_c} w_\beta U^{e+\beta} = \sum_{(\xi, \eta) \in \mathcal{P}_c} \frac{w_{\xi+\eta}}{N(\xi+\eta)} U^{e+\xi+\eta}. \quad (5)$$

The linear fibre-sum map $Q : k^{\mathcal{P}_c} \rightarrow k^{\mathcal{W}_c}$ has exact star relations and an integral reference section. Over $\mathbb{Q}$ its counting-norm canonical section is $J(c)_x = c_{q(x)}/N(q(x))$. The full inverse retains QA and every nonreference coefficient of $A - JQA$. Its quotient norm is $\sum |c_\beta|^2/N(\beta)$.

Proof. Each interval in (4) is nonempty since $|\beta_i| \leq f_i + b_i$; its bounds are exactly the two summand bounds. The residue product proves surjectivity and (5). For the linear assertion, on each fibre sum-zero vectors are exactly its star relations, and equal coefficients minimize the counting norm. The relation primitive is the list of nonreference coefficients. All inverse compositions follow by substitution. $\square$

The original uniform norm on $k^{\mathcal{W}_c}$ is *not* this native quotient norm. Transporting it requires diagonal cost $N(\beta)$ on each refined occurrence. This choice is forced among fibre-permutation-invariant diagonal metrics: cost h gives quotient cost $h/N(\beta)$. Every change of metric is consequently explicit.

For finite surjections $P \rightarrow W \rightarrow Z$, put $m_w = |P_w|$ and $n_z = \sum_{w \mapsto z} m_w$. With the inherited metric $1/m_w$, the second canonical inverse gives $m_w c_z/n_z$ at w , and the first gives c_z/n_z at every original occurrence. This is exactly the direct inverse. Resetting the intermediate metric changes it. For three copies of $[-1, 1]$, total exponent zero has seven lifts. Direct coefficients are all $1/7$, squared norm $1/7$; resetting yields four $1/6$ and three $1/9$, squared norm $4/27$. The nonzero boundary discarded by resetting has squared norm

$$4/27 - 1/7 = 1/189. \quad (6)$$

The same object keeps failures and successful denominators. For any pair put $x = v(\xi)$, $w = v(\eta)$, $\rho = x - t_c w^{-1}$ and $\beta = \xi + \eta$. The original gates obey

$$g_M = u + a \equiv aw\rho, \quad g_E = 4u + 1 \equiv pw\rho \pmod{R}.$$

Both factors are units, so $D = R/\gcd(R, g_c) = R/\gcd(R, \rho)$. Formula (5) extends to *all* pairs by adding Z^{D-1} to each monomial and using $\mathcal{W}_c = B_e$. All original prime-power defects are retained. Multiplication by the same unit gives the explicit affine coefficient-system conjugacy of [4], with inverse multiplication by its modular inverse.

For a matched pair, set $d_0 = \gcd(a, u)$, $(h, r, s) = (d_0^2/u, u/d_0, a/d_0)$. On M let $\lambda = (r + s)/R$ and take $(a, phs\lambda, phr\lambda)$. On E let $\kappa = (pr + s)/R$ and take $(a, hsk, phr\kappa)$. The gate makes the quotient a positive integer; clearing reciprocals proves $4/p$. The ordered inverse is $u = pa^2/(RY - pa)$ on M and $u = a^2/(RY - pa)$ on E. Divisor valuations recover β , and (4) recovers every split lift. The cohomology rank counts residues, not pairs or original divisor states; those two projections of $\mathcal{P}_c$ remain separate.

Exact examples and nonclaims. At $p = 2521, a = 636, R = 23$, split $(2, 1, 1) = (1, 1, 1) + (1, 0, 0)$. There are five matched pairs, four channel/residue classes, and three original divisor states. Four pairs have $N = 2$, one has $N = 1$, so the original mass is $4/2 + 1 = 3$. The harmonic Gram is $\text{diag}(2, 2, 3/2, 2)$, determinant 12. At the empty shell $p = 241, a = 64, R = 15$, split $6 = 3 + 3$. The full two-channel complex has dimensions $(12, 28, 16)$ and is exact. The raw two-leg kernel has dimension twelve, all relations, while the all-ones source norm remains 28. A supported zero cohomology fibre is not an absent source. Neither example is new prime coverage.

The general proofs above are self-contained finite algebra. The delivered standard-library checker verifies actual differentials, Gram determinants, coarsening primitives, the full marked examples, all small map pairs, all small overlapping splits and a bounded arithmetic scan. It does not prove universal ES or a zeta spectral identification. No historical priority or Lean certification is claimed.

References

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- [3] The Clankers (2026). *Canonical quotient volumes, trace control and an endpoint-budget criterion*, §§1,5. Same repository/revision, `workbenches/tau-toda-volume-formal/RESEARCH_NOTE.md`.
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